Jordan Watkins, PhD

Full-stack game programmer & project lead — C++ / SDL / OpenGL, shipped on Steam

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Co-lead programmer and lead puzzle designer on *Bean and Nothingness*, a logic-puzzle game rated 98% positive from 198 Steam reviews. I work across the whole stack of a game: engine and state systems, editor tooling shipped to players, gameplay and entity logic, custom audio, and puzzle design — and I coordinated a distributed remote team from prototype to release. Specializing in challenging indie puzzle-games. PhD mathematician who reasons precisely about state, correctness, and complex systems, with a decade of production service and database work alongside.

GAME DEVELOPMENT EXPERIENCE

Co-Developer — *Bean and Nothingness* · RBOR Games — 2020–2021 · shipped Dec 2021

store.steampowered.com/app/1706090/Bean_and_Nothingness/

Tile-based logic-puzzle game · 98% positive from 198 Steam reviews · 180+ puzzles · Steam Workshop support

- Co-lead programmer (C++, SDL, OpenGL); deeply familiar with nearly the full codebase. Grew out of prototype work begun in 2012
- Designed and implemented a replay / undo / state-saving system requiring precise modeling of game state across arbitrary action sequences
- Built a GUI level editor used internally and shipped publicly via Steam Workshop
- Implemented a custom sound mixer and contributed to the core game loop and entity logic architecture
- Lead puzzle designer: created 100+ puzzles, edited all ~200, and owned difficulty progression across the whole game
- Coordinated a distributed remote team from prototype through release — scoping, sequencing content, and driving the project to ship

SYSTEMS & BACKEND EXPERIENCE

Architect & Maintainer — Necrobot · Open source — 2015–Present

github.com/incnone/necrobot

Discord bots managing competitive speedrunning races and league seasons in *Crypt of the NecroDancer*, in continuous community use for 10+ years

- Architected one codebase that runs two distinct production bots (Necrobot and Condorbot) — ~150 modules across 16 packages with a documented, acyclic package dependency order
- Built an internal command framework separating input parsing and user-facing error handling from data manipulation, over an object layer mapped to MySQL tables
- Async Python (asyncio, aiohttp, discord.py) with MySQL persistence, Google Sheets integration for standings, and timezone-aware match scheduling
- Race ratings built on TrueSkill (Bayesian), adapted to the race format of the game
- Lead architect and maintainer across 10+ years, coordinating a small contributor team; MIT-licensed with admin, code, and database documentation

Competitive Rating & Matchmaking Systems · CoNDOR · open source — multiple seasons, through 2022

github.com/incnone/condor_ratings

Bayesian rating model, outcome forecasting, and ILP matchmaking for a multi-league season (Python, C++, MySQL, PuLP)

- Custom Bayesian paired-comparison model (Gaussian priors + binomial evidence) fit by MAP estimation with hand-derived gradients and a hand-written optimizer; prior variance chosen by held-out log-likelihood; heavy-tailed link function with its tail weight fit to past results
- Best-of-3 / best-of-5 outcome forecasts from jointly normal ratings with covariance — exposed by patching a C++ Bayesian Elo engine
- Weekly matchmaking as an integer linear program maximizing predicted-outcome entropy under degree constraints

SKILLS

Languages	C++ (primary), Python, SQL
Engine & graphics	SDL, OpenGL, custom audio mixing, replay/state systems, entity logic, core game loop
Tools	GUI level editor authoring, Steam Workshop, Git, Visual Studio
Design	Puzzle design and difficulty progression across a full commercial game

Project leadership	Scoping and sequencing content, coordinating distributed remote teams, driving releases, long-term maintenance
Backend	Async Python services, MySQL, rating and matchmaking algorithms
Foundations	PhD mathematics — geometry, linear algebra, probability

EDUCATION

Ph.D. in Mathematics · University of Michigan 2013

B.A. in Mathematics and Physics · Knox College 2007

Career note: independent study and freelance tutoring 2013–2020, during part of which I carried family caregiving responsibilities.